# Analyzing and Visualizing Data

## AWS Academy Data Engineering

# Introduction

## Analyzing and Visualizing Data

# Module objectives

This module prepares you to do the following:

- List factors to consider when selecting analysis and visualization tools.
- Compare available AWS tools and services for data analysis and visualization.
- Determine the appropriate AWS tools and services to analyze and visualize data based on influencing factors (business needs, data characteristics, and access to data).

# Module overview

## Presentation sections

- Considering factors that influence tool selection
- Comparing AWS tools and services
- Selecting tools for a gaming analytics use case

## Lab

- Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

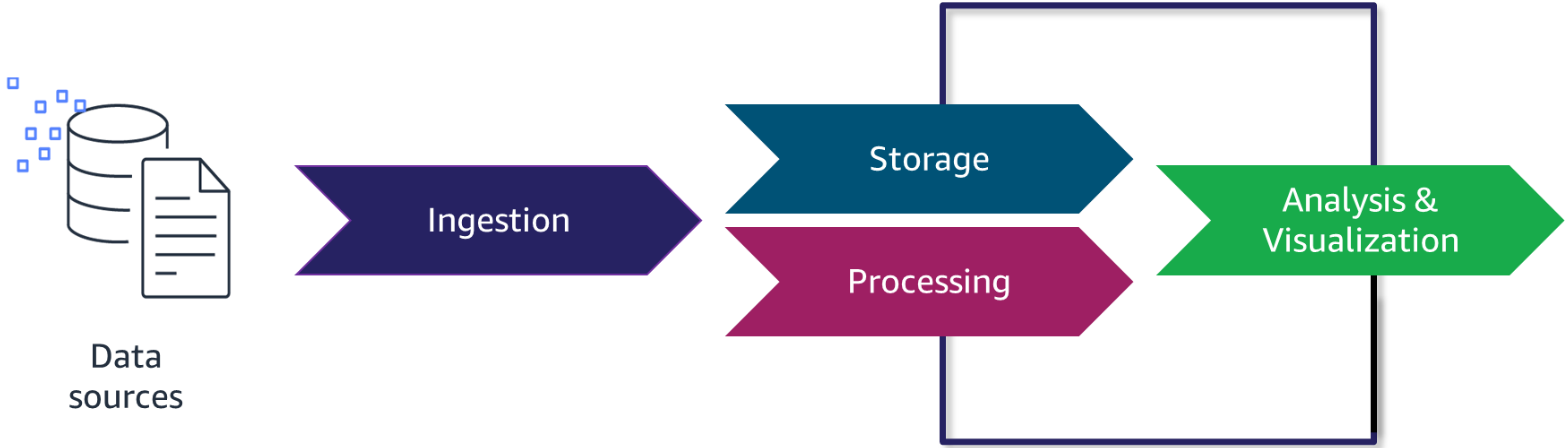
## Knowledge checks

- Online knowledge check
- Sample exam question

# Considering factors that influence tool selection

## Analyzing and Visualizing Data

# The simplified data pipeline



# Factors to consider when selecting tools

## Business needs

Fully understand the business needs to be able to determine which data analyses and visualizations are needed to help develop insights.

## Data characteristics

Understand the type and quality of the data, and how often it is updated and processed.

## Access to data

Consider the data pipeline and who needs access to analyze and visualize data.

# Business needs

- Which analyses are needed to help develop insights?
- What insights can be pulled from the data?
- What visualization would illustrate the insights?
- Does the consumer need to generate a report or interact with a dashboard?

# Business needs: Granularity of insight

| Industry | Detailed Level | Aggregate Level |
|---|---|---|
| **Finance** | A finance manager wants information (such as revenue, costs, and profit margins) about their line of business. | A CFO wants similar metrics at an aggregate level across all lines of businesses. They want to be able to drill down to any line of business. |
| **Marketing** | A marketing manager wants to know about the number of leads, opportunities, and closed deals within an area (such as a postal code or city). | A CMO is interested in related metrics but at a broader level (such as a state or region). |
| **Sales** | A sales manager is focused on their sales pipeline and wants to know how long it takes to close an opportunity. They want to assess how many opportunities are needed to achieve quota targets. | A VP of sales wants similar information at an aggregate level. They want to be able to drill down to a sales representative or sales territory. |

# Business needs: Visualizing insights

### KPIs

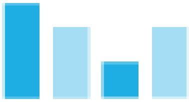
Show performance in a particular area or function.

### Relationships

Establish or prove whether a relationship exists between two or more variables.

### Comparisons

Show or examine how different variables change over time, or provide a static snapshot of how different variables compare.



### Distributions

Show how your data is distributed over certain intervals (based on clustering or grouping of data).

### Compositions

Highlight the various elements that make up your data—its composition.

# Data characteristics

- How much data is there?
- At what speed and volume does it arrive?
- How frequently is it updated?
- How quickly is it processed?
- What type of data is it?

# Data characteristics: Examples of data types

## Volume and velocity

- **Historical analysis:** Visualize a year's worth of sales data. Users can drill down by region and salesperson.
- **Streaming Internet of Things (IoT) data:** Visualize the real-time error rates of sensors in a factory.

## Variety and veracity

- **Structured data:** A relational database is queried to report on customer service tickets that were submitted in a specific period.
- **Unstructured data:** Sentiment analysis is performed on customer service emails.

## Value

- A business analyst uses **periodic reports** to showcase and report results to leadership.
- A DevOps engineer uses **self-service dashboards** to monitor and analyze performance in real time.

# Data characteristics: Two fraud detection use cases

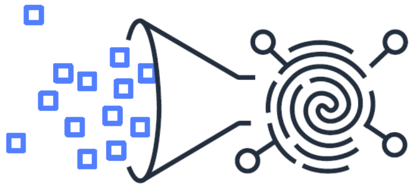
| Data characteristic considerations | Use case 1: Rule-Based (Batch Pipeline) | Use case 2: ML in Real Time (Streaming Pipeline) |
|---|---|---|
| How much data is there? | Millions of transactions (kilobytes to terabytes) | Millions of transactions (bytes to megabytes) |
| At what speed and volume is it arriving? | In predefined intervals (minutes to multiple days) | In real time (milliseconds to seconds) |
| How quickly is it processed? | Minutes to hours | Milliseconds to seconds |
| What type of data is it? | Structured and semistructured | Unstructured and semistructured data |
| What value do insights from the data provide? | Historical reporting of fraud cases<br>Reactive approach | Ability to detect fraud in real time<br>Proactive approach |

# Access to data

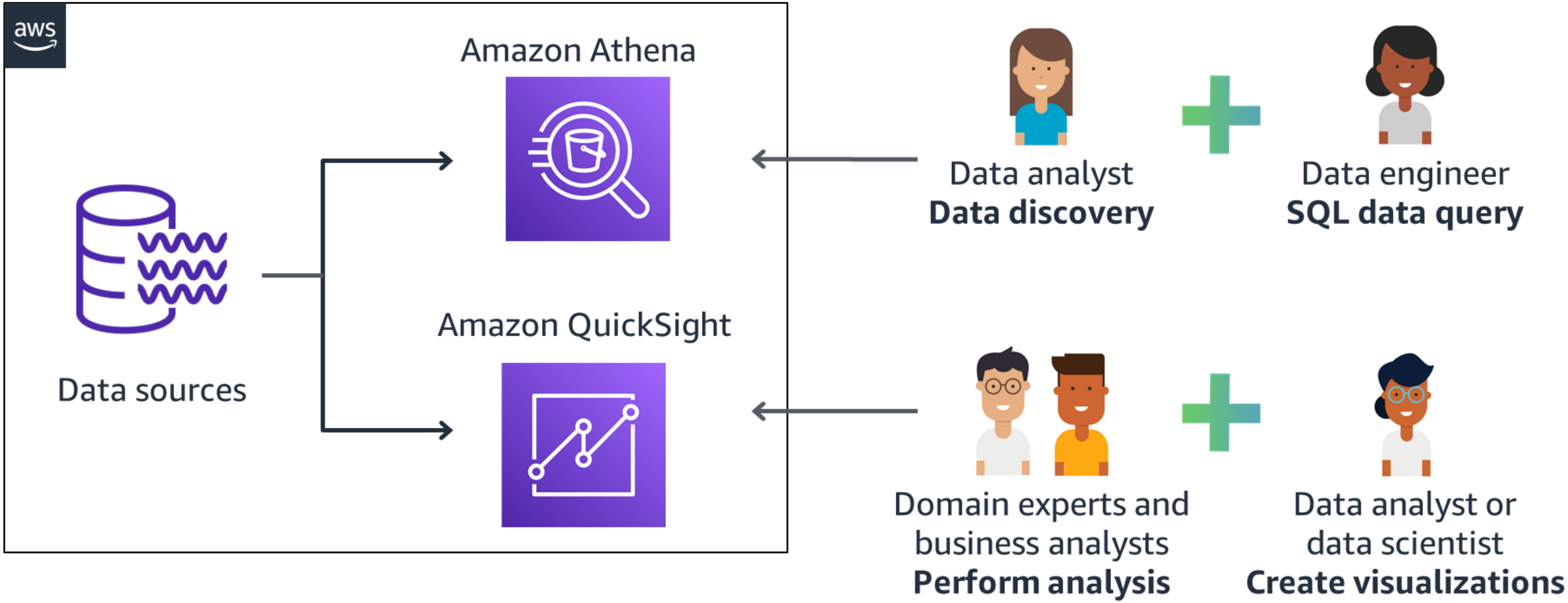
- Where does the data come from?
- Will the data need to be combined from multiple sources?
- Who needs access to the data and at what level? Who can access the tools?

# Access to data: **Consider authorization level based on role**

- A user’s authorization to access data depends on their role in the organization.
- A business analyst or manager might be authorized to read the output that data engineers or data analysts create but not delete or update it.
- Follow the principle of least privilege. Give users the least amount of access and responsibility that are needed to complete their duties.



# Access to data: Consider the functions of handling data



# Key takeaways: Considering factors that influence tool selection

- When you select analysis and visualization tools, consider the business needs, data characteristics, and access to data.
- Consider the granularity and format of the insights based on business needs.
- Consider the volume, velocity, variety, veracity, and value of your data.
- Consider the functions of individuals who will access, analyze, and visualize the data.

# Comparing AWS tools and services

## Analyzing and Visualizing Data

# AWS services in the data pipeline

**Ingestion**

- **Store raw data:** Amazon Simple Storage Service (Amazon S3)
- **Collect data:** Amazon Kinesis Data Streams and Amazon Data Firehose
- **Migrate data:** AWS DataSync, AWS Database Migration Service (AWS DMS), Amazon AppFlow, and AWS Glue

**Storage**

**Processing**

- **Store aggregated data:** Amazon S3, Amazon Redshift, Amazon Relational Database Service (Amazon RDS), Amazon OpenSearch Service, and Amazon SageMaker Data Wrangler
- **Process data:** Amazon Managed Service for Apache Flink, Amazon EMR, AWS Glue, SageMaker Data Wrangler

**Analysis and visualization**

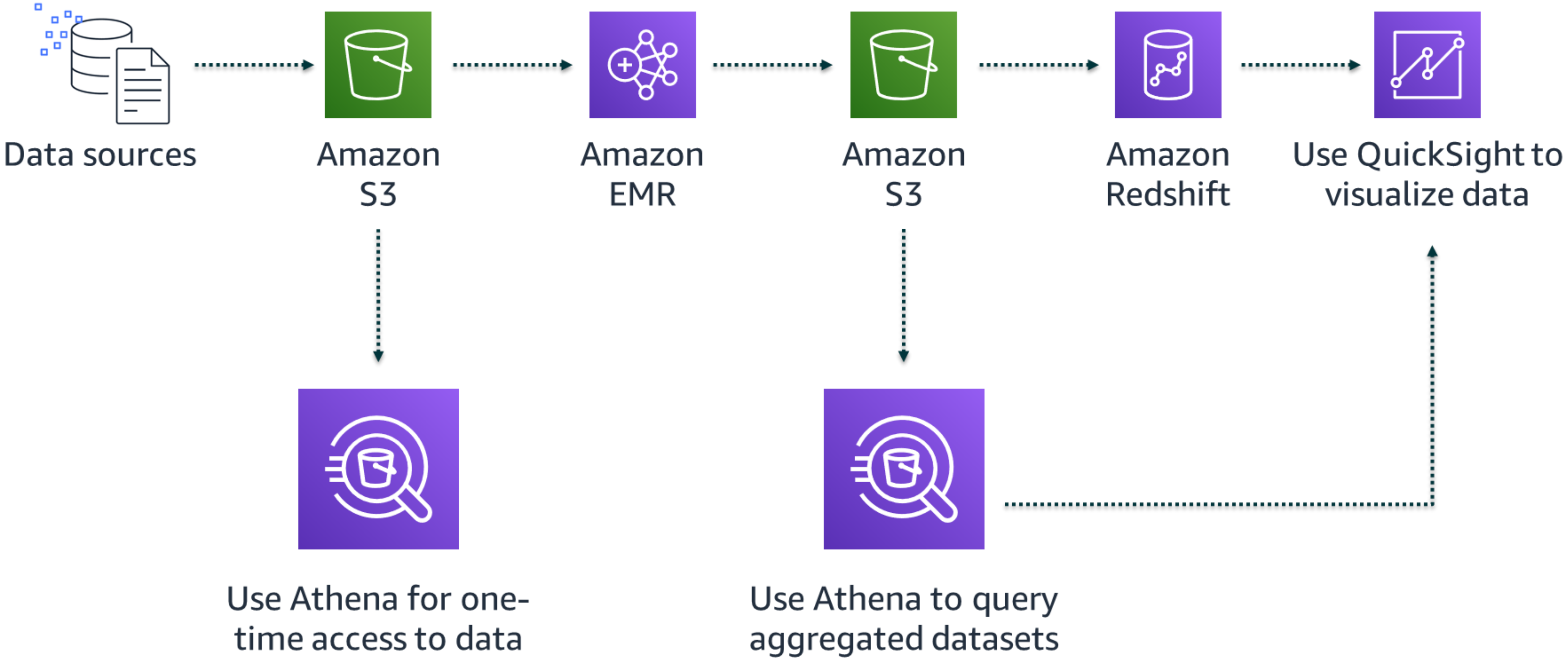
- **Visualize and query:** Athena, QuickSight, and OpenSearch Service

# Amazon Athena

Amazon Athena is an interactive query service that provides the ability to use SQL to analyze data in Amazon S3. Athena includes the following features:

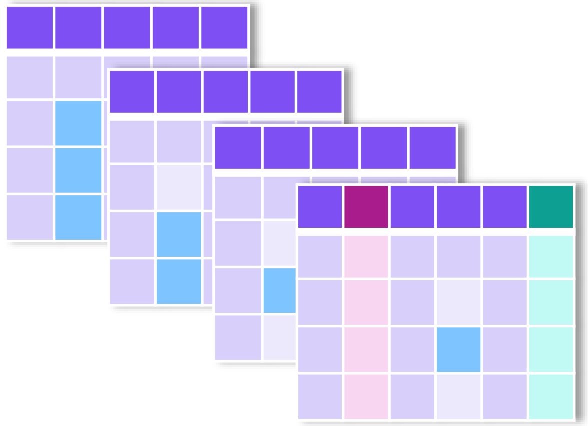
- Is serverless
- Provides the ability to combine data from multiple data sources
- Can be used for one-time queries
- Can be used from your favorite business intelligence (BI) tools (such as QuickSight)
- Can update data stored in Amazon S3 with Apache Iceberg integration

# Athena: One-time querying for data in Amazon S3



# Athena: Capability to update data in Amazon S3

- Users can use Athena to insert, update, and delete data that is stored in Amazon S3 (with the Apache Iceberg integration).
- Users can also track data versions automatically.
- The Apache Iceberg integration provides a way for continuous ingestion and updates.

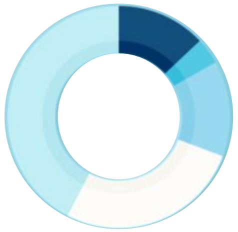


# Amazon QuickSight

- Is a cloud-scale BI service to deliver easy-to-understand insights
- Connects to data in the cloud and combines data from many different sources
- Gives decision-makers the opportunity to explore and interpret information in an interactive visual environment
- Provides forecasting visualization capabilities
- Provides ability to ask questions using natural language with QuickSight Q feature

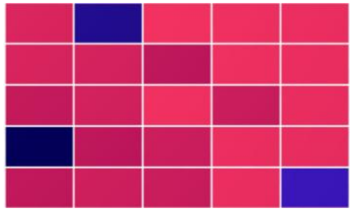
# QuickSight example use case

**To visualize the sentiments, phrases, and tweets for a specific topic in QuickSight, you will view the following:**

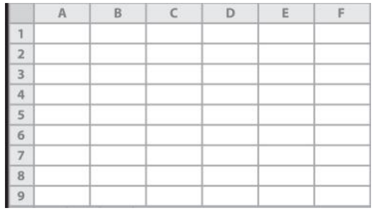


A donut chart with the overall sentiment for that specific topic sentiment


A word cloud of the phrases in the most dominant topic



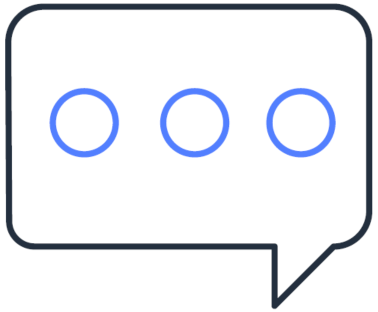
A heat map of tweets that are associated with the dominant topics



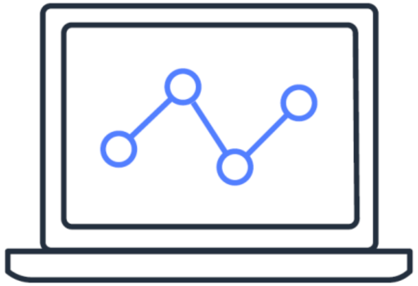

A tabular view of related tweets

# QuickSight Q: Ask questions using natural language

When you need a visualization that isn't already in the dashboard, instead of waiting for the BI team to get back to you, use QuickSight Q to get immediate responses.



**Example:** You could say "Show me last year's weekly sales in California," and Q would provide an answer in a few seconds.



- Is a BI capability that's powered by ML
- Uses natural language processing
- Doesn't require you to build pre-defined data models or dashboards

# Amazon OpenSearch Service

- This managed service helps you deploy, operate, and scale OpenSearch clusters in the AWS Cloud.
- Use this open-source search and analytics engine for use cases such as the following:
  - Log analytics
  - Real-time application monitoring
  - Clickstream analytics

Amazon OpenSearch Service is integrated with visualization tools including OpenSearch Dashboards and Kibana.

# Example use case for OpenSearch Dashboards

**Situation:** Analyze and visualize support calls

**Solution:**

- Use Amazon S3, Amazon Transcribe, and Amazon Comprehend to get full transcripts of calls, keywords from the transcripts, and an overall sentiment of each call.
- Use OpenSearch Service and OpenSearch Dashboards to search and visualize the data:
  - A pie chart of the overall sentiment (positive, negative, neutral)
  - A bar chart of the keywords that are mentioned in the support calls
  - A histogram of when and how often the calls were made

# Comparison of AWS services for data analysis and visualization

## Athena

- Interactive analysis using SQL
- Analyze data directly
- Start querying data instantly
- Serverless

## QuickSight

- Dashboards and visualizations
- Build visualizations and dashboards for business analytics
- Serverless

## OpenSearch Service

- Operational analytics
- Search, explore, filter, aggregate, and visualize data in near real time
- Fully managed service

# Key takeaways: Comparing AWS tools and services

- AWS tools and services that are commonly used to query and visualize data include Athena, QuickSight, and OpenSearch Service.
- Athena is used for interactive analysis with SQL.
- Decision-makers can use QuickSight to interact with data visually and get insight quickly.
- OpenSearch Service is used for operational analytics to visualize data in near real time.

# Selecting tools for a gaming analytics use case

## Analyzing and Visualizing Data

# Applying what you have learned in a use case for gaming analytics

**Factors that influence selection of analysis and visualization tools**

- Business needs
- Data characteristics
- Access to data

**AWS tools and services for analysis and visualization**

- Athena
- QuickSight
- OpenSearch Service

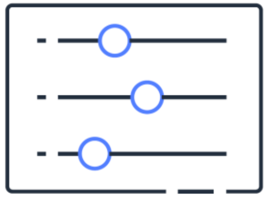
**Selecting tools for a gaming analytics use case**

- Solutions based on a particular use case
- Solutions based on personas in the use case

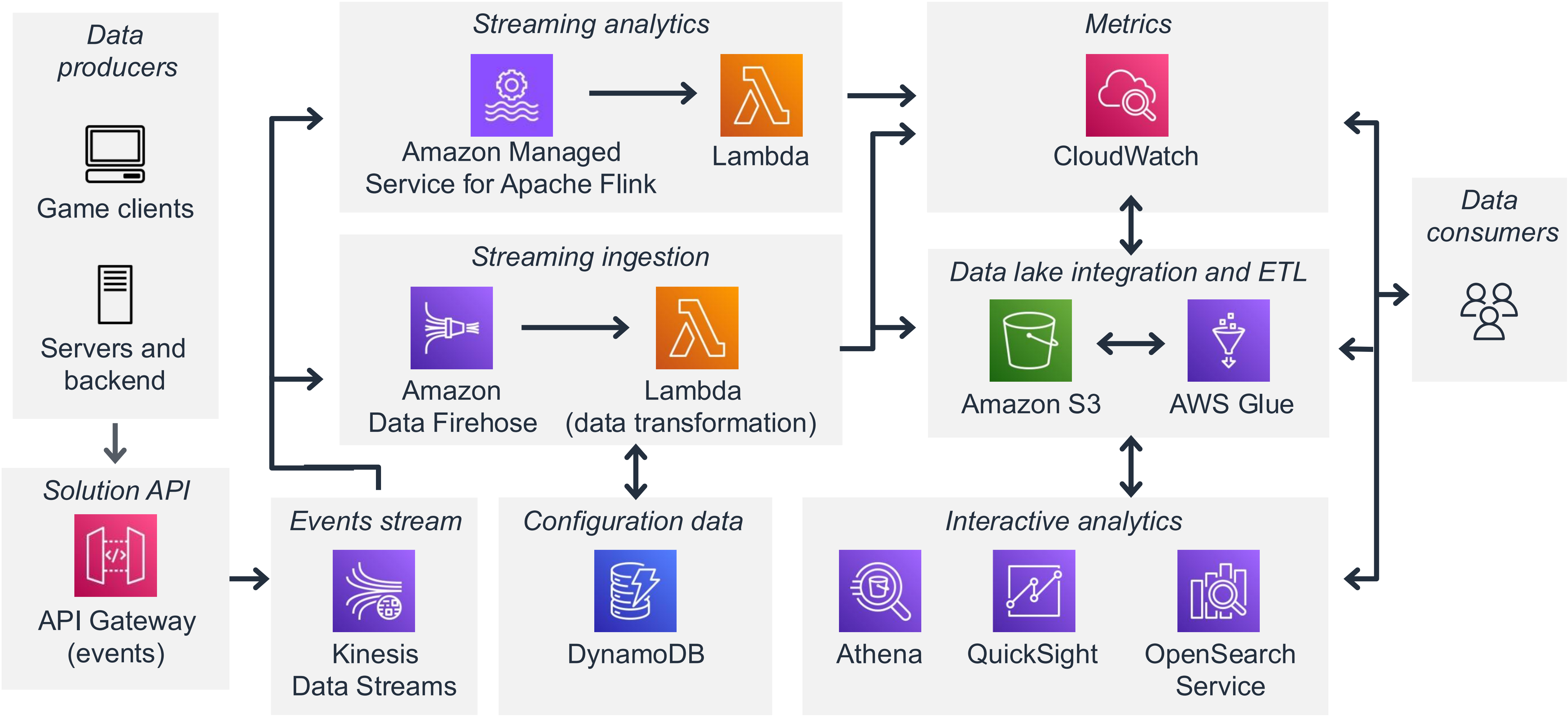
# Three personas in this use case for gaming analytics

How do different personas in a gaming company use AWS tools and services within different stages of the gaming analytics pipeline?

- **Analyst:** Explore and analyze player data
- **Business user:** Showcase and report results to leadership
- **DevOps engineer:** Monitor and analyze performance in real time

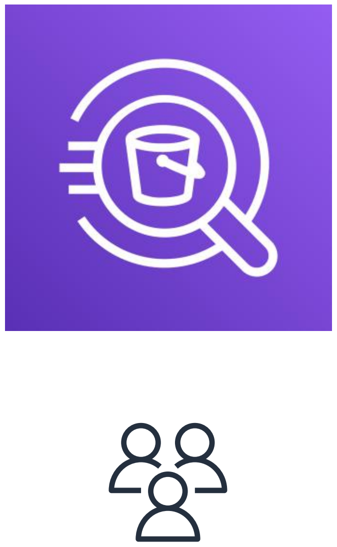


# Gaming analytics pipeline



# Using Athena to explore and analyze player data

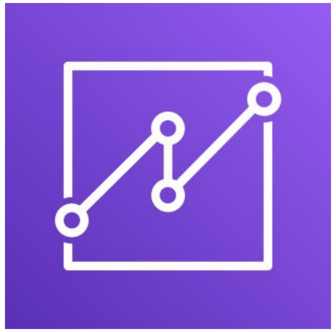
- **Business need:** Generate insights by querying daily aggregates of player usage data
- **Data characteristics:** Batch data for financial and geographical usage patterns
- **Data accessed:** Player purchase history, play history, and geographic information



Data analysts

# Using QuickSight to showcase and report results

- **Business need:** Visualize KPIs, such as average revenue per user, average revenue per paying user, retention rate, and conversion rate and forecasting
- **Data characteristics:** Combined data from multiple sources and aggregated to a high granularity level
- **Data accessed:** Player purchase history, play history, and geographic information



Business users

# Using OpenSearch Service to monitor and analyze performance in real time

- **Business need:** Monitor health and performance, and analyze performance for predictive load balancing
- **Data characteristics:** Large volumes of streaming telemetry data and server logs, including structured and unstructured data
- **Data accessed:** Access logs and performance data for game servers

DevOps engineers

# Key takeaways: Selecting tools for a gaming analytics use case

- This use case showcased the granularity of visualized insights:
  - Daily batch aggregates of client usage patterns (Athena)
  - Consolidated aggregate KPIs for leadership (QuickSight)
  - Continuous health and performance monitoring (OpenSearch Service)
- Keep the influencing factors in mind when you select AWS tools and services. Multiple solutions exist to meet the business needs of data analysis and visualization.

# Lab: Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

# Lab introduction:

## Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

- In this lab, you will analyze user activity and access patterns for a university bookstore website by using streaming log data.
- You will build an index for CloudWatch log events in OpenSearch Service.
- Then, you will use OpenSearch Dashboards to visualize the data:
  - Create a pie chart to show the operating systems and browsers that visitors use to consume the website.
  - Create a heat map to show how users are referred to product pages.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform during this lab.

# Debrief: Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards

- In this lab, you had to log in to OpenSearch Service with credentials other than your AWS credentials.
  - Which service managed user authentication to OpenSearch Service?
  - Which service managed user authorization?
  - Why were AWS services needed as part of the architecture in this lab?
- What is the purpose of index patterns in OpenSearch?
- What was the purpose of the Lambda function named **os-demo-lambda-function**?
- What was the purpose of the **agent.json** file on the Apache web server?
- Why was CloudWatch needed as part of the overall solution?

# Module wrap-up

## Analyzing and Visualizing Data

# Module summary

This module prepared you to do the following:

- List factors to consider when selecting analysis and visualization tools.
- Compare available AWS tools and services for data analysis and visualization.
- Determine the appropriate AWS tools and services to analyze and visualize data based on influencing factors (business needs, data characteristics, and access to data).

# Module knowledge check

- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

# Sample exam question

A company's web application produces 250 GB of clickstream data per day, which is stored in Amazon S3. The company wants to use the data to find out how quickly the application's webpages loaded during the last month. They want to be able to compare month-to-month data to gain insights. Which data analysis and visualization solution would provide these capabilities while also minimizing the cost and complexity?

Identify the key words and phrases before continuing.

**The following are the key words and phrases:**

- 250 GB of **clickstream data** per day
- **How quickly** the application's webpages loaded during the **last month**
- **Compare month-to-month data** to gain insights
- **Minimizing the cost and complexity**

# Sample exam question: Response choices

A company's web application produces 250 GB of **clickstream data** per day, which is stored in Amazon S3. The company wants to use the data to find out **how quickly** the application's webpages loaded during the **last month**. They want to be able to **compare month-to-month** data to gain insights. Which data analysis and visualization solution would provide these capabilities while also **minimizing the cost and complexity**?

| Choice | Response |
|---|---|
| A | Use Amazon OpenSearch Service to analyze and OpenSearch Dashboards to visualize the data. |
| B | Use Apache Spark to analyze the data and Amazon EMR Hive to visualize the data. |
| C | Use Amazon Managed Service for Apache Flink to analyze the data and Microsoft Excel to visualize the data. |
| D | Use Amazon Athena to analyze the data and Amazon QuickSight to visualize the data. |

# Sample exam question: Answer

The correct answer is D.

| Choice | Response |
| --- | --- |
| D | Use Amazon Athena to analyze the data and Amazon QuickSight to visualize the data. |

# Thank you

Corrections, feedback, or other questions?
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